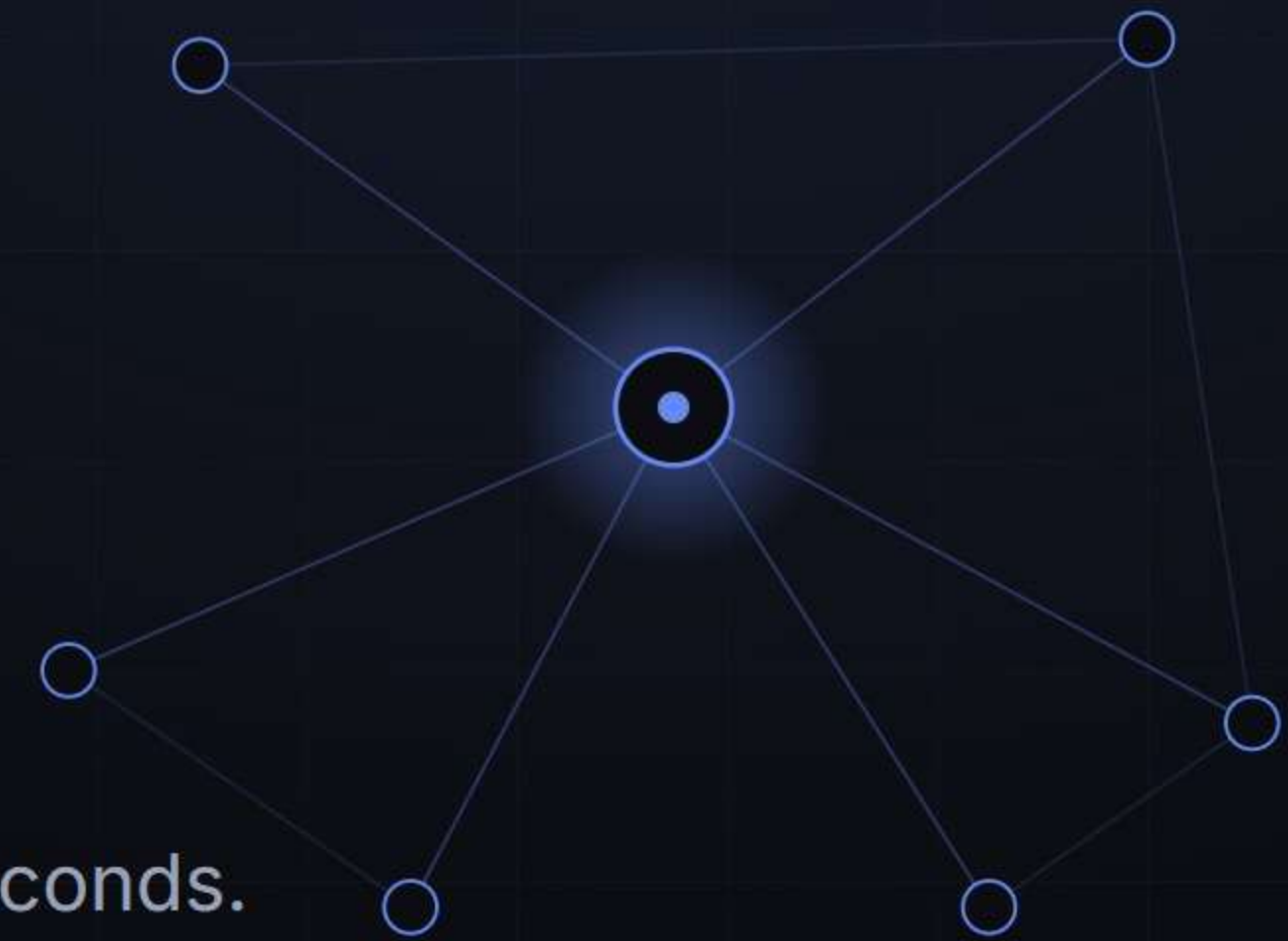


PROBLEM STATEMENT 8 · UNIFIED ASSET & OPERATIONS BRAIN

The Unified Asset & Operations **Brain.**

Turning a plant's fragmented knowledge into a cited, computed answer — in seconds.



ASSET-INTENSIVE INDUSTRY, TODAY

The knowledge exists. It's just scattered.

TIME LOST

35%

of a professional's day spent
searching for information.

McKINSEY • 2024

FRAGMENTATION

7–12

disconnected document systems in
the average large Indian plant.

NASSCOM • EY

DOWNTIME

18–22%

of unplanned downtime driven by
knowledge fragmentation.

BIS RESEARCH

KNOWLEDGE CLIFF

~25%

of India's experienced industrial
engineers retire within the decade.

INDUSTRY DATA

It's not a file-management problem. It's a **safety, quality, and efficiency** problem — and it compounds.

WHY SEARCH ALONE FAILS

Search finds documents. It *can't reason* about a pipe.

THE SITUATION

A pump's vibration cannot be diagnosed without knowing a **valve two nodes upstream** is cavitating.

Sanket makes the plant's **P&ID topology** the substrate the AI reasons over — not a keyword filter.

CAUSAL CHAIN · ONE HOP UPSTREAM



TOPOLOGY → connections are the substrate, not the metadata.

ONE BRAIN · FIVE CAPABILITIES · ALL LIVE

Not five tools. One brain that does five things.

01



Universal Ingestion & Knowledge Graph

Any document — PDFs, emails, scans, work orders — parsed into a connected graph of assets, people and regulations.

02



Expert Knowledge Copilot

Natural-language questions answered from the whole corpus — with source citations and computed confidence.

03

FLAGSHIP



Maintenance & Root-Cause Analysis

Graph-topology reasoning traces a symptom to its true root cause upstream — not just similar text.

04



Quality & Regulatory Compliance

OISD, Factory Act, PESO, ISO clauses mapped to each asset's evidence — gaps flagged, audit packs generated.

05



Lessons-Learned & Failure Intelligence

A retiring expert's undocumented know-how becomes a permanent, queryable node in the graph.

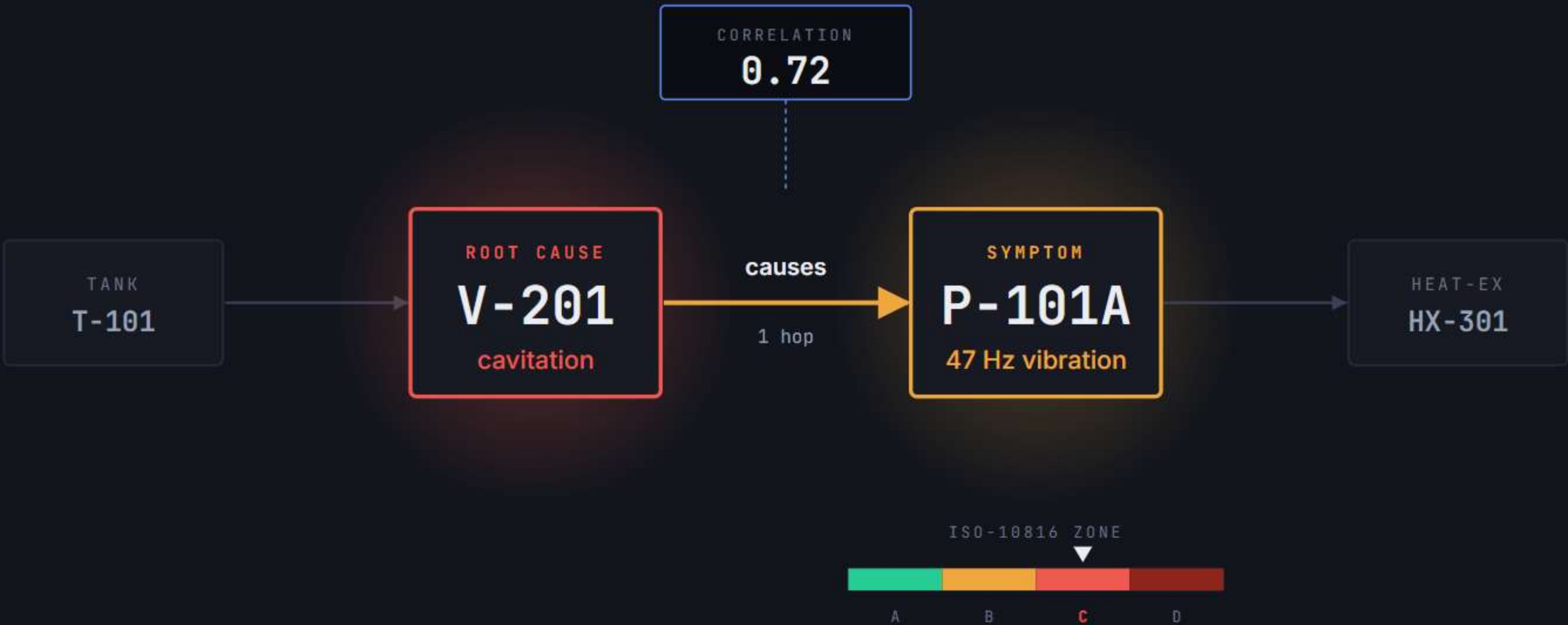
● ALL FIVE CAPABILITIES SHIPPED & DEPLOYED IN THE LIVE PLATFORM

QUERY

> P-101A pump 47 Hz vibration – why?

• ANSWERED in 0.7s
vs ~4h manual

P&ID TOPOLOGY · CAUSAL PATH LIT UP



SANKET · COMPUTED ANSWER

97%

COMPUTED CONFIDENCE

Root cause: V-201 control-valve cavitation, one hop upstream of P-101A.

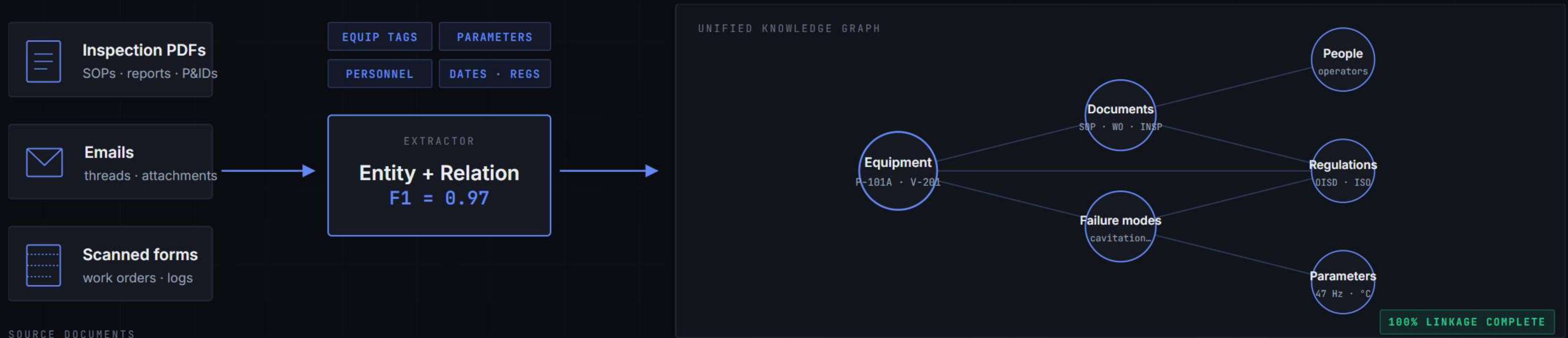
CITED TO REAL DOCS

EVIDENCE-COMPUTED CONFIDENCE

SECONDS · VS ~4h MANUAL

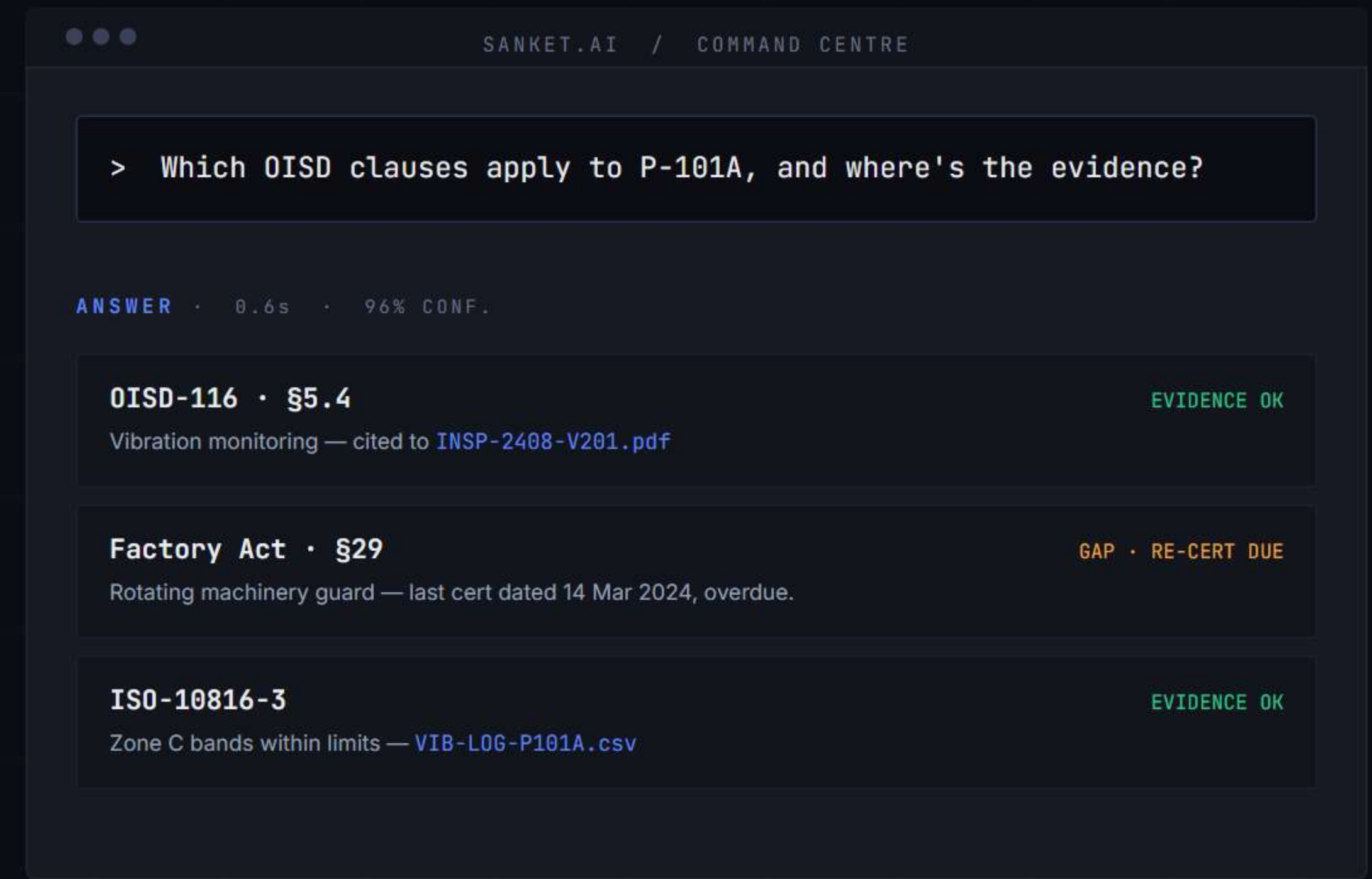
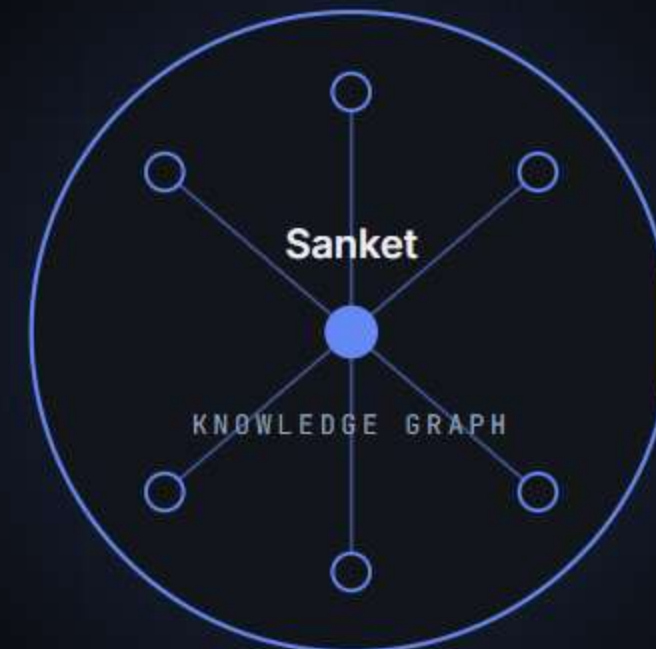
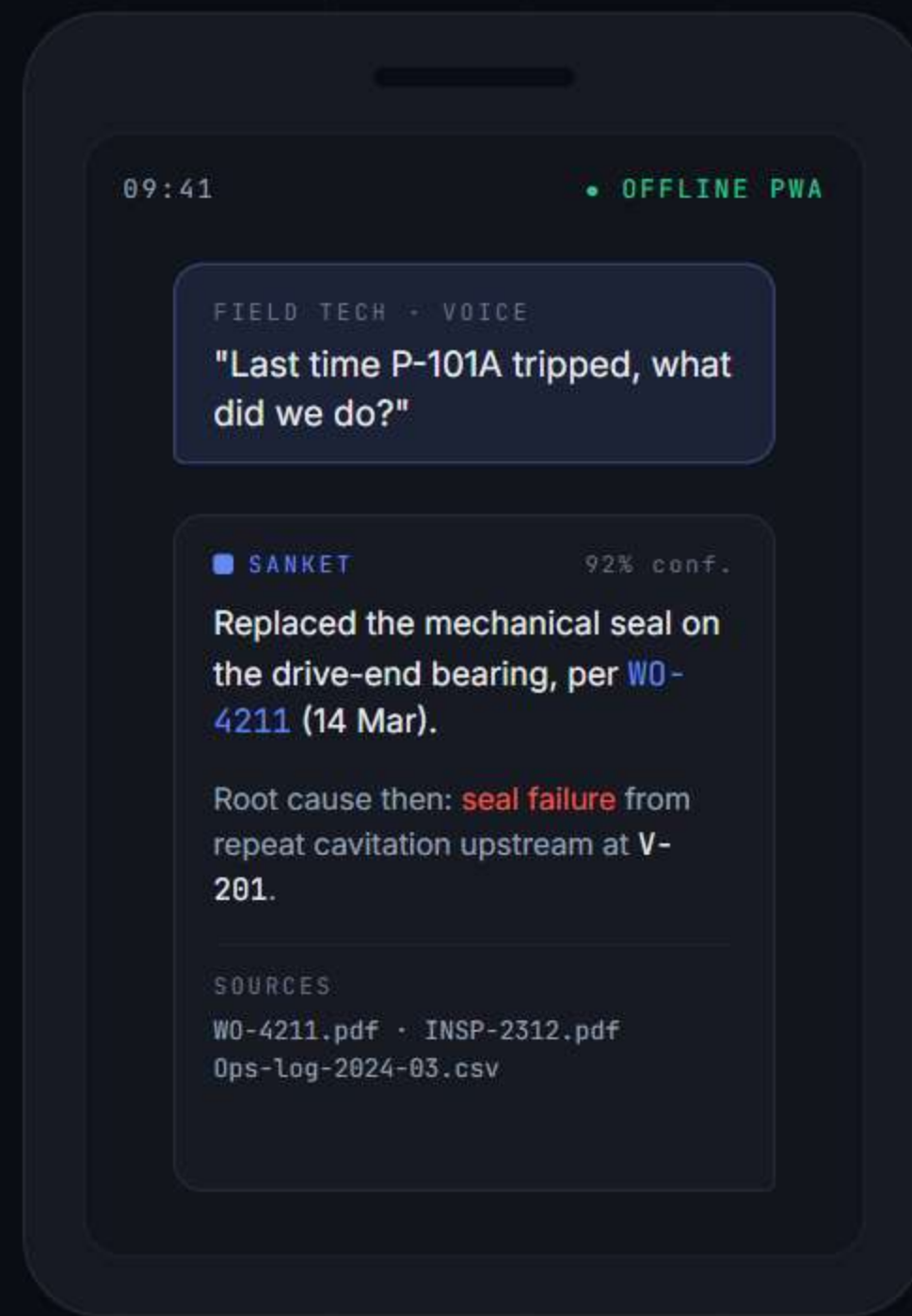
SOURCES: INSP-2408-V201.pdf · W0-4419.pdf
VIB-LOG-P101A.csv · ISO-10816-3

Any document. One connected brain.



Five node types — equipment, documents, failure modes, people, regulations — linked into **one queryable brain**.

Ask anything. Get a cited answer.



Same brain. Field technician on a phone, engineer at a desk — **every answer cited, every confidence computed.**

Continuous compliance. Permanent expertise.

PILLAR 4 · COMPLIANCE

Regulatory clauses,
mapped to real evidence.

OISD

FACTORY ACT

PESO

ISO

CLAUSE

OISD-116 §5.4

ASSET

P-101A

EVIDENCE

Insp. log · WO

▲ Gaps flagged automatically. ▶ Audit packs generated in one click.

PILLAR 5 · THE KNOWLEDGE CLIFF

Undocumented know-how
becomes a graph node.

**Retiring expert**

30 yrs · P-101A

CAPTURE

STRUCTURED
INTERVIEW

LINK

KG NODE



SAMPLE QUERY

> "How did Ravi handle the 2019 P-101A trip?" → answered

Every number computed. Every citation verified.

01



Confidence is scored, not asserted.

From evidence, not the model:

- doc count & recency
- severity & ISO correlation
- confirmed graph path

02



Hallucinated citations are detected & flagged.

Every citation is resolved back to a real document,
real page, real span — or it is refused.

● FABRICATION = 0

03



Deterministic engine. Reproducible & testable.

The same inputs produce the same graph — every
time.

33

AUTOMATED TESTS
ALL PASSING

Trust is engineered, not promised — the model's role is bounded, and the plant's data is the source of truth.

The numbers, from our benchmark harness.

ANSWER ACCURACY

100%

on the benchmark question set.

CITATION FAITHFULNESS

100%

zero fabricated sources.

ENTITY EXTRACTION

97%F1 across equipment,
params, dates.

KG LINKAGE

100%

every extracted entity resolved.

TIME-TO-ANSWER

<1s

vs ~4h manual search.

HEAD-TO-HEAD · ROOT-CAUSE ACCURACY

Finds the true root cause — 100% vs 17%.

Sanket — graph topology

**100%**

Keyword search

**17%**

MID-SIZE PLANT · ANNUAL VALUE

~₹2.17 Cr

per year — labour saved + unplanned downtime avoided.

Grounded in the brief's own figures — **McKinsey 35%** time-lost, **BIS 18–22%** downtime-driver.

HOW IT ADDS UP

Search time reduction (35%)	+ ₹1.24 Cr
Unplanned downtime avoided	+ ₹0.78 Cr
Compliance & audit efficiency	+ ₹0.15 Cr

TOTAL / YEAR

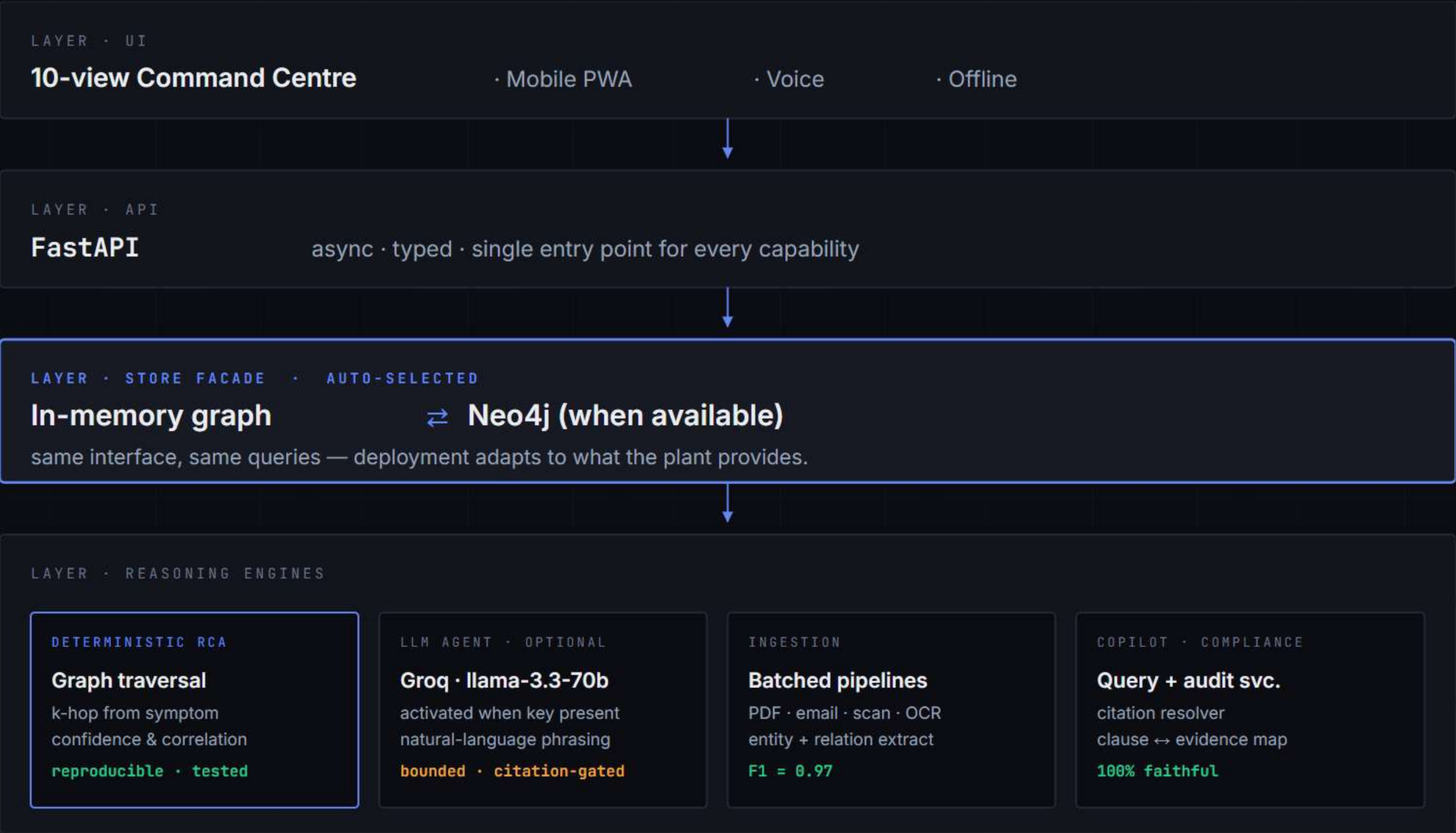
₹2.17 Cr

WHY THE VALUE SCALES

Reasoning cost is **bounded to the k-hop neighbourhood** around a query — independent of plant size.

A 10,000-tag plant answers as fast as a 500-tag one. Value grows linearly; cost does not.

Runs today with zero external services. Scales tomorrow with Neo4j + Groq.



SCALES BY DESIGN

01 · ZERO-DEPS DEFAULT

Runs with no external services — in-memory graph + deterministic engine.

02 · BOUNDED REASONING

k-hop neighbourhood queries — a **10,000-tag** plant answers as fast as 500.

03 · GRACEFUL UPGRADE

Same code, richer backend — Neo4j and a Groq LLM agent light up when they're available.

Every rubric weight — earned.

CRITERION	WEIGHT	SANKET'S EDGE	EVIDENCE
Innovation	25%	Graph-topology reasoning — topology is the substrate, not a filter.	V-201 → P-101A · 1-hop RCA
Business Impact	25%	₹2.17 Cr / year / plant — grounded in the brief's own data.	McKinsey 35% · BIS 18–22%
Technical Excellence	20%	Computed confidence + verified citations + deterministic engine.	33 tests · 100% faithful · 97% F1
Scalability	15%	Neo4j-optional store facade; k-hop bounded reasoning cost.	10,000-tag plant · same latency
User Experience	15%	10-view command centre, mobile, voice, offline PWA.	Same brain, every surface

THE HOOK, RESTATED

Search gives you documents. Sanket gives you **answers**.

Cited, computed, and connected
across every system a plant runs on.

The unified asset and operations brain.

END · Q&A

Thank you.
Questions?